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T1 — Biological Molecules

1.1 Water

Water: A polar molecule consisting of two hydrogen atoms covalently bonded to one oxygen atom (H_2O). The oxygen atom is more electronegative, creating an uneven distribution of charge (a dipole). This polarity enables hydrogen bonding between the slightly positive hydrogen of one water molecule and the slightly negative oxygen of another.

Properties of Water

Water is essential for life due to several biologically important properties, all arising from hydrogen bonding between water molecules:

Property	Explanation	Biological Importance
High specific heat capacity (4.18 J g^{-1} K^{-1})	Hydrogen bonds require significant energy input to break, so water absorbs a lot of heat before its temperature rises	Provides a thermally stable environment for aquatic organisms and for biochemical reactions inside cells; helps mammals maintain constant body temperature
High latent heat of vaporisation	Many hydrogen bonds must be broken for water molecules to escape as vapour	Evaporation of sweat cools the body effectively; transpiration cools leaves
Cohesion and surface tension	Hydrogen bonds hold water molecules together (cohesion); at the surface, molecules are pulled inward creating tension	Cohesion allows water columns in xylem vessels to be pulled upward under tension; surface tension supports small organisms (e.g. pond skaters)

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Excellent solvent	Polar water molecules are attracted to ions and other polar molecules, surrounding them and pulling them into solution	Transport medium for metabolites, ions, and waste products in blood plasma and cytoplasm; all biochemical reactions occur in aqueous solution
Maximum density at 4 degrees C (ice floats)	Below 4 degrees C, hydrogen bonds form an open crystalline lattice in ice, making it less dense than liquid water	Ice forms an insulating layer on lakes and rivers in winter, allowing aquatic organisms to survive beneath; prevents water bodies from freezing solid
Adhesion	Water molecules are attracted to other polar surfaces (e.g. cellulose in xylem walls)	Helps draw water up xylem walls (capillary action); water spreads across mesophyll cell surfaces for gas exchange

Exam Tip: When explaining water's properties, always link the molecular explanation (hydrogen bonds, polarity) to a specific biological function. Simply stating "water is a good solvent" without explaining why (polarity attracts ions/polar molecules) and giving a biological context (e.g. dissolving glucose for transport in blood) will not gain full marks.

1.2 Carbohydrates

Carbohydrate: An organic molecule containing carbon, hydrogen, and oxygen, with the general formula $C_n(H_2O)_n$. They function as energy sources, energy stores, and structural components. Carbohydrates are classified as monosaccharides (single sugars), disaccharides (two sugars joined), or polysaccharides (many sugars joined).

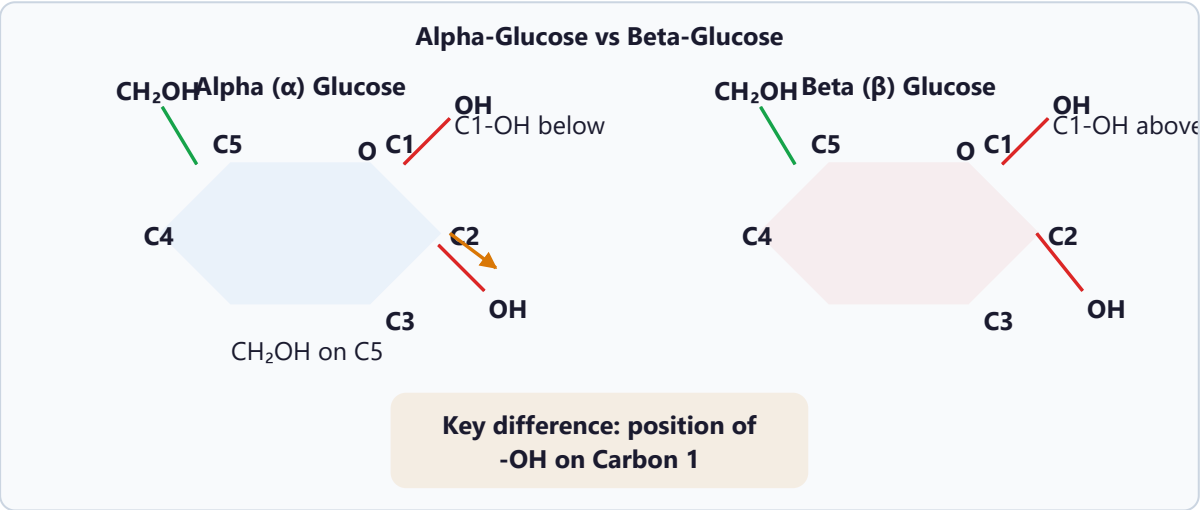
Monosaccharides

Monosaccharides are the simplest sugars and the monomers of carbohydrates. They are classified by the number of carbon atoms they contain:

- **Trioses (3C):** e.g. glyceraldehyde — key intermediates in respiration (glycolysis) and photosynthesis (Calvin cycle)

- **Pentoses (5C):** e.g. ribose (found in RNA and ATP) and deoxyribose (found in DNA). Deoxyribose lacks an oxygen atom on carbon-2 compared to ribose.
- **Hexoses (6C):** e.g. glucose (alpha and beta isomers), fructose, galactose. Glucose is the most important respiratory substrate in living organisms.

Alpha-glucose and beta-glucose are structural isomers (same molecular formula $C_6H_{12}O_6$) that differ only in the position of the hydroxyl (-OH) group on carbon-1. In alpha-glucose, the -OH on C1 points downward (below the plane of the ring); in beta-glucose, it points upward. This seemingly small difference has profound consequences for the polysaccharides they form.



Disaccharides

Two monosaccharides join via a **condensation reaction** (a molecule of water is released) forming a **glycosidic bond**. The reverse reaction is **hydrolysis** — the addition of water breaks the glycosidic bond, separating the monomers. Hydrolysis occurs during digestion and is catalysed by specific enzymes.

$$\text{Monosaccharide} + \text{Monosaccharide} \rightarrow \text{Disaccharide} + H_2O \text{ (condensation)}$$

$$\text{Disaccharide} + H_2O \rightarrow \text{Monosaccharide} + \text{Monosaccharide (hydrolysis)}$$

Disaccharide	Monomers	Bond	Found in
Maltose	Glucose + Glucose	α-1,4 glycosidic	Germinating seeds (starch digestion intermediate)

Sucrose	Glucose + Fructose	α-1,2 glycosidic	Plants (transport sugar in phloem sap)
Lactose	Glucose + Galactose	β-1,4 glycosidic	Mammalian milk (energy source for infants)

Polysaccharides

Polysaccharides are long chains of monosaccharides joined by glycosidic bonds. They are not sweet, generally insoluble (making them ideal for storage), and do not affect water potential.

Starch is the main storage polysaccharide in plants, made entirely from alpha-glucose. It has two components:

- **Amylose (~20%):** An unbranched chain of alpha-glucose with α-1,4 glycosidic bonds. The chain coils into a compact helix, making it space-efficient for storage. Its compactness also makes it insoluble, so it does not affect the water potential of the cell.
- **Amylopectin (~80%):** A branched chain with α-1,4 bonds along the main chain and α-1,6 bonds at branch points (roughly every 24-30 glucose units). The many branch ends provide sites for rapid enzymatic hydrolysis, enabling quick glucose release when needed.

Glycogen is the main storage polysaccharide in animals, stored primarily in liver and muscle cells. It is structurally similar to amylopectin but even more highly branched (branch points every 8-12 glucose units). The very high degree of branching means many terminal glucose residues are available for simultaneous enzyme attack, enabling extremely rapid glucose mobilisation — essential during sudden increases in metabolic demand such as exercise.

Cellulose is a structural polysaccharide in plant cell walls, made from **beta-glucose**. Because the -OH on C1 is in the beta position, adjacent glucose molecules must be rotated 180 degrees relative to each other to form β-1,4 glycosidic bonds. This produces long, straight, unbranched chains. Multiple cellulose chains lie parallel to one another and are cross-linked by hydrogen bonds between the -OH groups, forming **microfibrils**. These microfibrils have very high tensile strength, allowing cellulose cell walls to withstand turgor pressure and provide structural support.

Exam Tip: Starch and glycogen are insoluble and compact — explain that this means they do not affect water potential by osmosis, making them ideal for storage. Also: cellulose's strength comes from the many hydrogen bonds *between* parallel chains (inter-chain), not from the glycosidic bonds

within individual chains. Examiners often test this distinction. Remember that humans cannot digest cellulose because we lack the enzyme cellulase to break beta-1,4 glycosidic bonds.

1.3 Lipids

Lipid: A diverse group of hydrophobic biological molecules that are insoluble in water but soluble in organic solvents (e.g. ethanol, acetone). They contain carbon, hydrogen, and oxygen, but with a much higher proportion of hydrogen relative to oxygen compared to carbohydrates. This means lipids contain more C-H bonds and therefore release more energy per gram when oxidised.

Triglycerides

A triglyceride consists of one **glycerol** molecule (a 3-carbon alcohol) bonded to three **fatty acid** chains by **ester bonds**, formed via three condensation reactions (releasing 3 molecules of water).

$$\text{Glycerol} + 3\ \text{Fatty Acids} \rightarrow \text{Triglyceride} + 3H_2O$$

Each fatty acid has a long hydrocarbon tail (typically 14-20 carbons) and a carboxyl (-COOH) head group. The hydrocarbon tails are non-polar and hydrophobic, which is why triglycerides are insoluble in water.

Functions of triglycerides:

- **Energy storage:** The high ratio of hydrogen to oxygen atoms means more C-H bonds per molecule, providing approximately 37 kJ/g when oxidised — more than double the energy yield of carbohydrates (~16 kJ/g). This makes fat an efficient long-term energy store.
- **Thermal insulation:** Subcutaneous adipose tissue (beneath the skin) reduces heat loss, particularly important in marine mammals such as whales and seals (blubber layer).
- **Protection:** Fat deposits surround and cushion vital organs (e.g. kidneys, heart).
- **Metabolic water:** Oxidation of fat produces more water per gram than any other respiratory substrate. Desert animals like camels metabolise fat reserves to produce water — the hump stores fat, not water.
- **Buoyancy:** Fat is less dense than water, contributing to buoyancy in aquatic organisms.
- **Electrical insulation:** The myelin sheath around nerve axons contains lipids and acts as an electrical insulator, enabling saltatory conduction.

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Saturated fatty acids: No C=C double bonds in the hydrocarbon tail; chains are straight and pack tightly together via van der Waals forces — solid at room temperature (e.g. animal fats like butter and lard). Diets high in saturated fat are linked to increased LDL cholesterol and cardiovascular disease risk.

Unsaturated fatty acids: One (monounsaturated) or more (polyunsaturated) C=C double bonds in the hydrocarbon tail; each double bond causes a kink/bend in the chain, preventing close packing — liquid at room temperature (e.g. plant oils like olive oil and sunflower oil, fish oils). Generally considered healthier for the cardiovascular system.

Phospholipids

Phospholipids have a glycerol backbone bonded to **two** fatty acid tails and one **phosphate head group** (which may carry additional polar groups). The phosphate head is **hydrophilic** (polar, interacts with water) while the fatty acid tails are **hydrophobic** (non-polar, repelled by water). This makes phospholipids **amphipathic**.

In an aqueous environment, phospholipids spontaneously arrange into a **bilayer**: the hydrophilic heads face outward toward the water, and the hydrophobic tails face inward, away from water. This phospholipid bilayer is the fundamental structural basis of all cell membranes. The hydrophobic interior acts as a barrier to most polar and charged molecules, controlling what enters and leaves the cell.

Cholesterol

Cholesterol is a lipid with a compact, rigid ring structure. It sits between phospholipid molecules in cell membranes, with its hydroxyl (-OH) group interacting with the phospholipid head and its hydrophobic rings and tail embedded in the hydrophobic core. Cholesterol has a dual role in membrane regulation:

- At **high temperatures**: cholesterol reduces membrane fluidity by restricting phospholipid movement, preventing the membrane from becoming too permeable.
- At **low temperatures**: cholesterol prevents phospholipids from packing too closely, maintaining fluidity and preventing the membrane from becoming rigid and brittle.

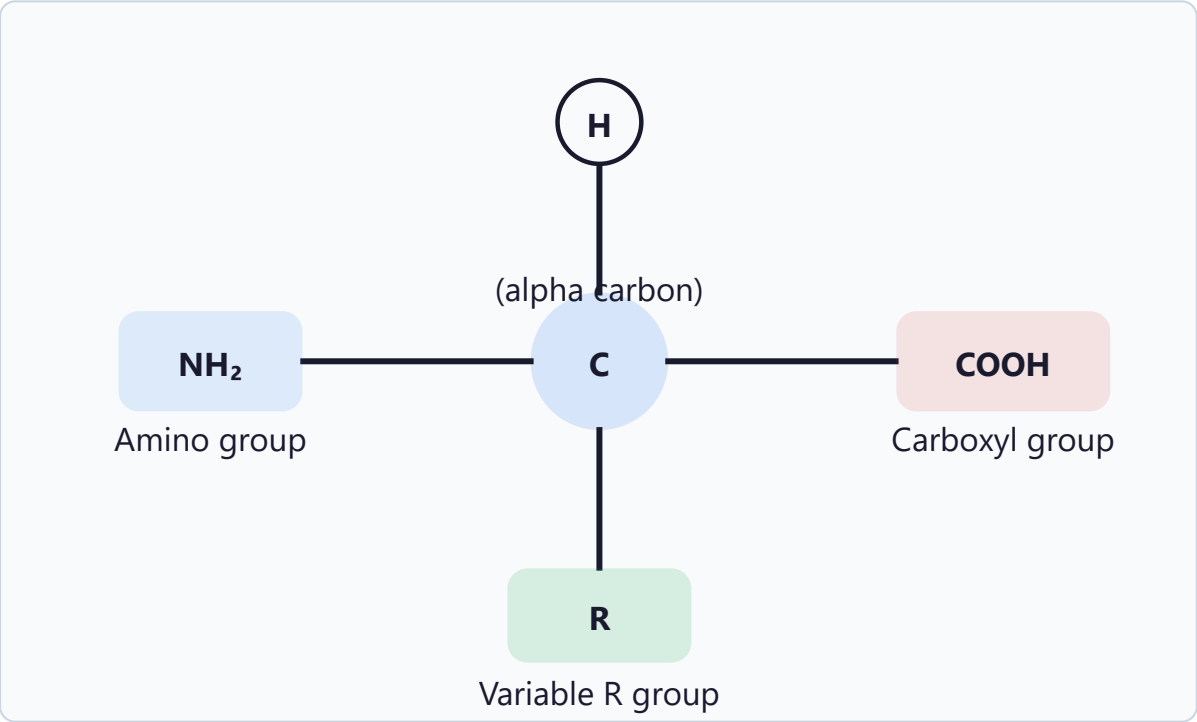
Cholesterol is also a precursor for steroid hormones (e.g. oestrogen, testosterone, cortisol) and vitamin D, and is a component of bile salts.

1.4 Proteins

Protein: A large biological polymer made of amino acid monomers joined by peptide bonds. Proteins carry out a vast range of functions: catalysis (enzymes), transport (haemoglobin, channel proteins), structural support (collagen, keratin), immunity (antibodies), signalling (hormones like insulin), movement (actin and myosin), and storage.

Amino Acid Structure

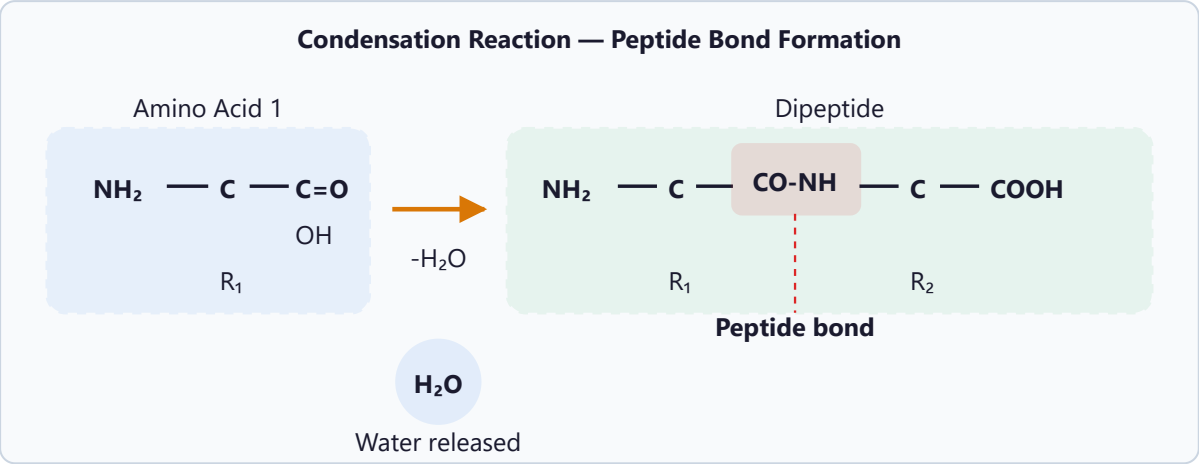
All 20 amino acids share a common structure: a central (alpha) carbon atom bonded to four groups: an amino group ($-NH_2$), a carboxyl group (-COOH), a hydrogen atom (-H), and a variable **R group** (side chain). The R group determines each amino acid's unique chemical properties — it may be non-polar (hydrophobic), polar (hydrophilic), positively charged, negatively charged, or contain special groups like sulfhydryl (-SH) in cysteine.



Peptide Bond Formation

Amino acids join by condensation reactions between the amino group of one amino acid and the carboxyl group of the next, forming a **peptide bond** (-CO-NH-) and releasing water. Two amino acids form a dipeptide; many amino acids form a polypeptide. Peptide bonds are broken by hydrolysis (addition of water).

$$\text{Amino Acid} + \text{Amino Acid} \rightarrow \text{Dipeptide} + H_2O \text{ (condensation / peptide bond formation)}$$



Levels of Protein Structure

Level	Description	Bonds Involved
Primary	The specific linear sequence of amino acids in a polypeptide chain, determined by the gene. Even a single amino acid change can alter the protein's function (e.g. sickle cell haemoglobin).	Peptide bonds between amino acids
Secondary	Regular local folding patterns: **alpha-helices** (coiled like a spring) and **beta-pleated sheets** (folded in a zigzag). These form due to hydrogen bonding between the C=O and N-H groups of the polypeptide backbone (not the R groups).	Hydrogen bonds between backbone groups
Tertiary	The overall 3D shape of a single polypeptide chain, formed by interactions between R groups. This is the level that determines the protein's specific function, including the shape of enzyme active sites and antibody binding sites.	Hydrogen bonds, ionic bonds, disulfide bridges (covalent, between cysteine residues), hydrophobic interactions (non-polar R groups cluster in the interior, away from water)
Quaternary	Two or more polypeptide chains (subunits) associated together to form a functional protein. May also include non-protein prosthetic groups.	Same as tertiary, between subunits. E.g. haemoglobin has 4 subunits (2 alpha, 2

		beta) each with a haem prosthetic group.

Globular vs Fibrous Proteins

Feature	Globular Proteins	Fibrous Proteins
Shape	Compact, roughly spherical, tightly folded	Long, thin, repetitive secondary structure
Solubility	Soluble (hydrophilic R groups on outer surface)	Insoluble (hydrophobic R groups throughout)
Function	Metabolically active: enzymes, haemoglobin, antibodies, hormones	Structural roles: tensile strength, support, protection
Examples	Enzymes, haemoglobin, insulin, antibodies	Collagen (skin, bone, tendons), keratin (hair, nails), elastin

Enzymes

Enzyme: A biological catalyst — a globular protein that increases the rate of a specific metabolic reaction by lowering the activation energy (E_a), without being consumed or permanently altered in the reaction. Each enzyme has a uniquely shaped **active site** — a small region (usually a cleft or pocket) on its surface where the substrate binds.

Lock and Key Model (Fischer, 1894): The substrate has an exactly complementary shape to the enzyme's active site. The substrate fits precisely into the active site like a key into a lock, forming an enzyme-substrate (ES) complex. This model explains enzyme specificity but does not account for the flexibility of enzymes.

Induced Fit Model (Koshland, 1958 — more widely accepted): The active site is not a rigid, perfectly complementary shape. Instead, when the substrate approaches and begins to bind, the active site **changes shape** slightly, moulding itself more closely around the substrate. This conformational change places strain on bonds within the substrate (lowering activation energy), and also brings catalytic R groups into the optimal position. The induced fit model explains why some molecules that are similar in shape to the substrate can still be poor substrates — they do not induce the correct conformational change.

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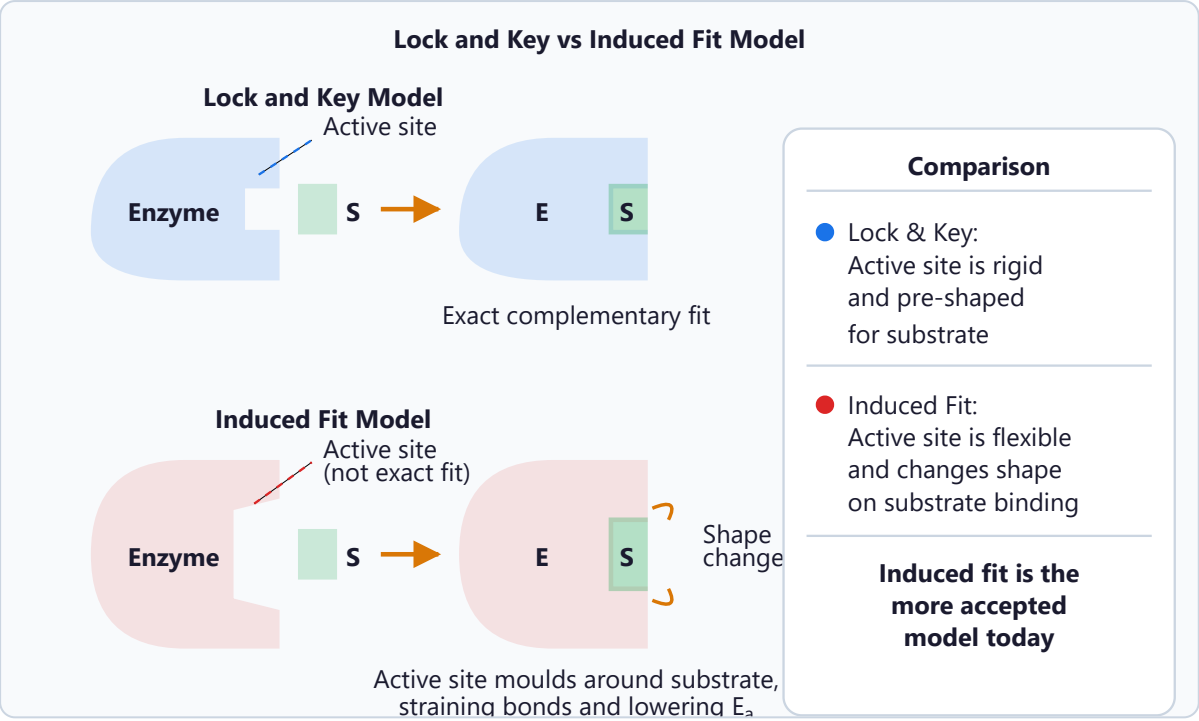

Factors Affecting Enzyme Activity

- **Temperature:** Increasing temperature provides more kinetic energy to enzyme and substrate molecules, increasing the frequency and energy of collisions, so the rate of ES complex formation increases. Rate roughly doubles for every 10 degrees C rise (Q_{10} rule) up to the **optimum temperature** (typically ~37 degrees C for human enzymes). Above the optimum, hydrogen bonds, ionic bonds, and hydrophobic interactions that maintain the tertiary structure begin to break. The active site loses its specific shape (**denaturation**) and the substrate can no longer bind. Activity drops rapidly to zero. Denaturation is usually permanent for most enzymes.
- **pH:** Changes in H^+ concentration alter the ionisation of R groups (amino and carboxyl groups become protonated or deprotonated), disrupting ionic bonds and hydrogen bonds that maintain the active site shape. Each enzyme has an optimum pH (e.g. pepsin ~pH 2 in the stomach, trypsin ~pH 8 in the duodenum). Extremes of pH cause denaturation.
- **Substrate concentration:** At low [S], rate increases proportionally as there are more substrate molecules to occupy active sites. Eventually, all active sites are occupied (enzyme saturation) and the rate plateaus at V_{max}. Increasing [S] beyond this point has no effect — enzyme concentration is now the limiting factor.
- **Enzyme concentration:** If substrate is in excess, increasing enzyme concentration increases the rate proportionally (more active sites available). If substrate becomes limiting, the rate plateaus.

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Enzyme Inhibitors

Type	Mechanism	Effect of Increasing [S]	Example
Competitive	Inhibitor has a similar shape to the substrate and binds directly to the **active site**, blocking substrate access. The inhibitor competes with substrate for the active site.	Can overcome inhibition by increasing [S] (outcompetes inhibitor). V_{max} unchanged but apparent K_m increased.	Malonate inhibits succinate dehydrogenase (competes with succinate in the Krebs cycle); statins compete at HMG-CoA reductase active site
Non-competitive	Inhibitor binds to an **allosteric site** (not the active site), causing a conformational change that distorts the active site shape. The substrate can still bind but the ES complex cannot form products, or forms them at a reduced rate.	Cannot overcome by increasing [S] because the inhibitor does not compete for the same site. V_{max} reduced but K_m unchanged.	Cyanide inhibits cytochrome c oxidase; heavy metals (Hg^{2+}, Pb^{2+}) bind to -SH groups; organophosphate nerve agents inhibit acetylcholinesterase

Exam Tip: Always distinguish between denaturation and inhibition. Denaturation involves permanent structural change due to broken bonds maintaining tertiary structure — it is irreversible (in most cases). Inhibition (especially competitive) is often reversible and does not permanently alter the enzyme's structure. Competitive inhibitors are useful drugs precisely because they are reversible and can be outcompeted. Also: do not say "the active site is destroyed" — say "the active site changes shape so the substrate can no longer bind."

1.5 Nucleic Acids

Nucleic Acid: A biological polymer made of nucleotide monomers. DNA (deoxyribonucleic acid) stores the genetic information that determines the sequence of amino acids in proteins. RNA

(ribonucleic acid) has several roles in protein synthesis, including carrying the genetic message (mRNA), transporting amino acids (tRNA), and forming part of ribosomes (rRNA).

Nucleotide Structure

A nucleotide consists of three components joined by condensation reactions:

- A **pentose sugar** — deoxyribose (in DNA, with -H on C2) or ribose (in RNA, with -OH on C2)
- A **phosphate group** — attached to C5 of the sugar
- A **nitrogenous base** — attached to C1 of the sugar. Purines (adenine, guanine) have a double-ring structure; pyrimidines (cytosine, thymine/uracil) have a single ring. A purine always pairs with a pyrimidine, maintaining a constant width of the double helix.

Nucleotides join by **phosphodiester bonds** — condensation reactions between the phosphate group on C5 of one nucleotide and the hydroxyl group on C3 of the next sugar. This creates a continuous **sugar-phosphate backbone** with directionality: one end has a free 5' phosphate (5' end) and the other has a free 3' hydroxyl (3' end).

DNA Structure

DNA is a double-stranded helix (Watson and Crick model, 1953, based on X-ray crystallography data from Franklin and Wilkins). Two antiparallel polynucleotide strands are held together by **hydrogen bonds between complementary base pairs**:

- **Adenine (A)** pairs with **Thymine (T)** via 2 hydrogen bonds
- **Cytosine (C)** pairs with **Guanine (G)** via 3 hydrogen bonds

Complementary base pairing means that the base sequence on one strand determines the sequence on the other — this is essential for accurate replication. The two strands run in opposite directions (**antiparallel**): one runs 5' to 3' and the other 3' to 5'. This is critical for replication (DNA polymerase only synthesises 5' to 3') and transcription.

The sugar-phosphate backbones form the outer "rails" of the helix while the bases face inward. The double helix is stabilised by: hydrogen bonds between bases (individually weak but collectively strong due to the vast number), hydrophobic stacking interactions between adjacent base pairs, and the sugar-phosphate backbone covalent bonds.

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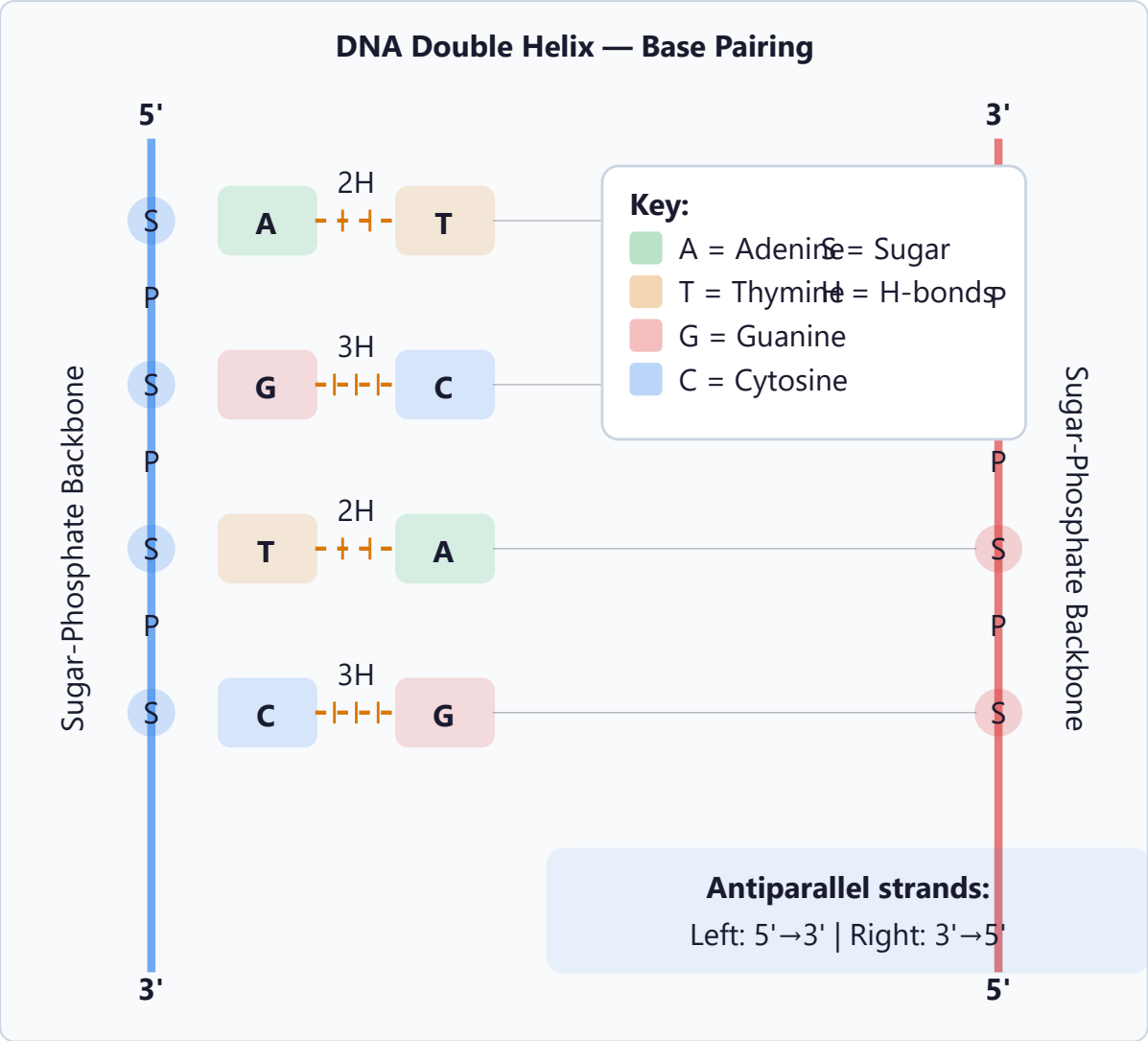

DNA vs RNA

Feature	DNA	RNA
Sugar	Deoxyribose (H on C2)	Ribose (OH on C2)
Bases	A, T, C, G	A, U, C, G (uracil replaces thymine)
Strands	Double-stranded (with base pairing)	Single-stranded (can fold into complex shapes via internal base pairing, e.g. tRNA cloverleaf)
Stability	Very stable (suited for long-term storage)	Less stable (suited for temporary roles)

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Function	Long-term storage of genetic information	Protein synthesis: mRNA (carries code), tRNA (carries amino acids), rRNA (forms ribosomes)
Location	Nucleus (main genome), mitochondria, chloroplasts	Nucleus (synthesised), cytoplasm (functions)

Exam Tip: Chargaff's rules: in DNA, the amount of A always equals T, and C always equals G (because of complementary base pairing). This does not apply to single-stranded RNA. If an exam question gives you the percentage of one base in DNA, you can calculate all others: e.g. if A = 30%, then T = 30%, and C + G = 40%, so C = G = 20%.

ATP (Adenosine Triphosphate)

ATP: A nucleotide derivative consisting of the base adenine, the sugar ribose, and three phosphate groups linked in a chain. It is the universal and immediate energy currency of all living cells, coupling energy-releasing reactions (exergonic) to energy-requiring reactions (endergonic).

$$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_\text{i} + \text{Energy } (\sim 30.6 \text{ kJ mol}^{-1}) \text{ — catalysed by ATPase}$$

$$\text{ADP} + \text{P}_\text{i} + \text{Energy} \rightarrow \text{ATP — catalysed by ATP synthase (in mitochondria/chloroplasts)}$$

Properties making ATP an ideal energy currency:

- **Small, manageable energy packets:** Each hydrolysis releases ~30.6 kJ mol^{-1} — enough to drive most cellular reactions without wasting energy as heat. Compare with glucose (~2870 kJ mol^{-1}) which would release far too much energy in one step.
- **Rapidly reformed:** The ATP-ADP cycle turns over extremely rapidly. A resting human uses and regenerates approximately their own body weight in ATP per day. ATP is reformed by substrate-level phosphorylation (glycolysis, Krebs cycle) and oxidative phosphorylation/photophosphorylation.
- **Water soluble:** Its charged phosphate groups make it soluble, so it can be easily transported within cells.
- **Cannot cross membranes:** ATP cannot pass out of the cell, keeping the energy where it is needed.
- **Universal:** Used by every known living organism for the same range of cellular processes.

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- **Single hydrolysis step:** Energy is released in a single, fast reaction (unlike glucose which requires many enzyme-catalysed steps).

ATP provides energy for: active transport (pumping ions/molecules against concentration gradients), muscle contraction (myosin power stroke), protein synthesis (forming peptide bonds), DNA replication, nerve impulse transmission (Na^+/K^+ ATPase), cell division, bioluminescence, nitrogen fixation, and many more processes.

Exam Tip: ATP is NOT an energy store (glycogen and fat are energy stores). It is an immediate energy currency — think of it as the "cash" of the cell while glycogen is the "savings account." ATP is continuously made and used; there is never a large surplus. Do not confuse ATP with ADP — ATP has three phosphate groups and is the "charged" form.

1.6 Biochemical Tests

These qualitative tests are used to identify the presence of specific biological molecules in a sample. They are commonly examined and feature in required practicals.

Test	Reagent	Procedure	Positive Result	Tests for
Benedict's test	Benedict's reagent (alkaline copper sulfate solution)	Add equal volumes of sample and Benedict's; heat in water bath at 80 degrees C for 5 minutes	Colour change from blue to green → yellow → orange → brick red precipitate (semi-quantitative: more sugar = more red)	Reducing sugars (all monosaccharides and most disaccharides except sucrose)
Non-reducing sugar test	Dilute HCl, $NaHCO_3$, then Benedict's	First test with Benedict's (should be negative). Then boil sample with dilute HCl for 2 min to hydrolyse, neutralise with $NaHCO_3$, then re-test with Benedict's	Colour change after hydrolysis only	Non-reducing sugars (e.g. sucrose)

Iodine test	Iodine in potassium iodide (I_2/KI) solution	Add a few drops to sample at room temperature	Colour change from yellow-brown to blue-black	Starch (iodine molecules become trapped in the amylose helix)
Biuret test	NaOH then $CuSO_4$ (or combined Biuret reagent)	Add equal volume of NaOH, then add a few drops of dilute $CuSO_4$. Mix gently (no heating needed).	Colour change from pale blue to purple/lilac (Cu^{2+} forms a complex with peptide bonds)	Proteins (peptide bonds) — will not react with individual amino acids
Emulsion test	Ethanol then water	Dissolve/mix sample in ethanol, then add equal volume of water	White cloudy emulsion forms (tiny lipid droplets suspended in water refract light)	Lipids (fats and oils)

Exam Tip: For the non-reducing sugar test, you **must** first boil with dilute HCl to hydrolyse the glycosidic bond, then **neutralise** with $NaHCO_3$ before adding Benedict's reagent. Benedict's reagent is alkaline and does not work in acidic conditions — forgetting the neutralisation step is the most common error and will cost marks. Also: always state that you need to test with Benedict's *first* to confirm there are no reducing sugars present before doing the hydrolysis step.

1.7 Inorganic Ions

Inorganic ions play essential roles in biological processes despite being needed in small quantities:

- **Iron ions (Fe^{2+}):** Component of the haem prosthetic group in haemoglobin — reversibly binds oxygen for transport in red blood cells. Also found in cytochromes in the electron transport chain, where iron alternates between Fe^{2+} and Fe^{3+} as it transfers electrons. Iron deficiency causes anaemia (reduced oxygen-carrying capacity of blood).
- **Hydrogen ions (H^+):** Determine pH, affecting enzyme activity and protein structure. H^+ gradients across membranes (proton motive force) are crucial in chemiosmosis during oxidative phosphorylation and photophosphorylation — the flow of H^+ through ATP synthase drives ATP production.

- **Sodium ions (Na^+):** Essential for co-transport of glucose and amino acids in the ileum (sodium-glucose co-transporter); generation and propagation of nerve impulses (Na^+ influx during depolarisation); maintenance of blood pressure and fluid balance.
- **Potassium ions (K^+):** Essential for maintaining resting potential in neurones (K^+ leak channels); repolarisation during action potentials; stomatal opening in plants (K^+ influx into guard cells lowers water potential, causing water uptake and swelling).
- **Phosphate ions (PO_4^{3-}):** Component of nucleotides (DNA, RNA, ATP), phospholipids (cell membranes), and hydroxyapatite in bones and teeth. Phosphorylation/dephosphorylation of proteins is a key regulatory mechanism in cell signalling.
- **Calcium ions (Ca^{2+}):** Essential for muscle contraction (binds to troponin, exposing myosin binding sites on actin); blood clotting cascade; synaptic transmission (triggers vesicle fusion and neurotransmitter release); bone and tooth structure.

Exam Tip: Questions on inorganic ions typically ask you to link a specific ion to a specific biological process. Make sure you can give a precise role for each ion — vague answers like "ions are important for health" will not gain marks. The most commonly tested are iron in haemoglobin and hydrogen ions in chemiosmosis.

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